Active surface area by distance from the inlet — Phase 1, transmissivity σ = 0

Each cell: fractures whose centre lies in that 100 m bin, median over 5 seeds. Solid line: median dissolved fraction at the network PR_V minimum; dashed: at breakthrough.

Gypsum-only, P32 ×4 (n = 5)

Gypsum-only, P32 ×6 (n = 5)

Gypsum-only, P32 ×8 (n = 5)

Gypsum-only, P32 ×16 (n = 5)

Coupled gypsum–calcite, P32 ×4 (n = 5)

Coupled gypsum–calcite, P32 ×6 (n = 5)

Coupled gypsum–calcite, P32 ×8 (n = 5)

Coupled gypsum–calcite, P32 ×16 (n = 5)

Coupled ÷ gypsum-only, paired (n = 5)

Coupled ÷ gypsum-only, paired (n = 5)

Coupled ÷ gypsum-only, paired (n = 5)

Coupled ÷ gypsum-only, paired (n = 5)

fraction of the gypsum dissolved

0.0 0.2 0.4 0.6 0.8

0 200 400 600 800 1000

distance of fracture centre from inlet (m)

active surface area $d_Q/P32$ (Hyman), bin value / its start
(blue = flow concentrates, red = spreads)

3×10^0 2×10^0 10^0 6×10^{-1} 4×10^{-1}

coupled ÷ gypsum-only
(orange = coupled more spread)

2×10^0 10^0 6×10^{-1}

Active surface area by distance from the inlet — Phase 1, transmissivity σ = 0.5

Each cell: fractures whose centre lies in that 100 m bin, median over 5 seeds. Solid line: median dissolved fraction at the network PR_V minimum; dashed: at breakthrough.

Gypsum-only, P32 ×4 (n = 5)

Gypsum-only, P32 ×6 (n = 5)

Gypsum-only, P32 ×8 (n = 5)

Gypsum-only, P32 ×16 (n = 5)

Coupled gypsum–calcite, P32 ×4 (n = 5)

Coupled gypsum–calcite, P32 ×6 (n = 5)

Coupled gypsum–calcite, P32 ×8 (n = 5)

Coupled gypsum–calcite, P32 ×16 (n = 5)

Coupled ÷ gypsum-only, paired (n = 5)

Coupled ÷ gypsum-only, paired (n = 5)

Coupled ÷ gypsum-only, paired (n = 5)

Coupled ÷ gypsum-only, paired (n = 5)

fraction of the gypsum dissolved

0.0 0.2 0.4 0.6 0.8

0 200 400 600 800 1000

distance of fracture centre from inlet (m)

active surface area d_O/P32 (Hyman), bin value / its start
(blue = flow concentrates, red = spreads)

3×10^0 2×10^0 10^0 6×10^{-1} 4×10^{-1}

coupled ÷ gypsum-only
(orange = coupled more spread)

2×10^0 10^0 6×10^{-1}

Active surface area by distance from the inlet — Phase 1, transmissivity σ = 1

Each cell: fractures whose centre lies in that 100 m bin, median over 5 seeds. Solid line: median dissolved fraction at the network PR_V minimum; dashed: at breakthrough.

Gypsum-only, P32 ×4 (n = 5)

Gypsum-only, P32 ×6 (n = 5)

Gypsum-only, P32 ×8 (n = 5)

Gypsum-only, P32 ×16 (n = 5)

Coupled gypsum–calcite, P32 ×4 (n = 5)

Coupled gypsum–calcite, P32 ×6 (n = 5)

Coupled gypsum–calcite, P32 ×8 (n = 5)

Coupled gypsum–calcite, P32 ×16 (n = 5)

Coupled ÷ gypsum-only, paired (n = 5)

Coupled ÷ gypsum-only, paired (n = 5)

Coupled ÷ gypsum-only, paired (n = 5)

Coupled ÷ gypsum-only, paired (n = 5)

fraction of the gypsum dissolved

0.0

0.2

0.4

0.6

0.8

0

200

400

600

800

1000

distance of fracture centre from inlet (m)

active surface area d_O/P32 (Hyman), bin value / its start
(blue = flow concentrates, red = spreads)

3×10^{0}

2×10^{0}

10^{0}

6×10^{-1}

4×10^{-1}

coupled ÷ gypsum-only
(orange = coupled more spread)

2×10^{0}

10^{0}

6×10^{-1}

Participation ratio PR_V by distance from the inlet — Phase 1, transmissivity σ = 0

Each cell: fractures whose centre lies in that 100 m bin, median over 5 seeds. Solid line: median dissolved fraction at the network PR_V minimum; dashed: at breakthrough.

Gypsum-only, P32 ×4 (n = 5)

Gypsum-only, P32 ×6 (n = 5)

Gypsum-only, P32 ×8 (n = 5)

Gypsum-only, P32 ×16 (n = 5)

Coupled gypsum–calcite, P32 ×4 (n = 5)

Coupled gypsum–calcite, P32 ×6 (n = 5)

Coupled gypsum–calcite, P32 ×8 (n = 5)

Coupled gypsum–calcite, P32 ×16 (n = 5)

Coupled ÷ gypsum-only, paired (n = 5)

Coupled ÷ gypsum-only, paired (n = 5)

Coupled ÷ gypsum-only, paired (n = 5)

Coupled ÷ gypsum-only, paired (n = 5)

fraction of the gypsum dissolved

0.0
0.2
0.4
0.6
0.8

0
200
400
600
800
1000

distance of fracture centre from inlet (m)

participation ratio PR_V, bin value / its start
(blue = flow concentrates, red = spreads)

3×10^{0}
2×10^{0}
10^{0}
6×10^{-1}
4×10^{-1}

coupled ÷ gypsum-only
(orange = coupled more spread)

2×10^{0}
10^{0}
6×10^{-1}

Participation ratio PR_V by distance from the inlet — Phase 1, transmissivity σ = 0.5

Each cell: fractures whose centre lies in that 100 m bin, median over 5 seeds. Solid line: median dissolved fraction at the network PR_V minimum; dashed: at breakthrough.

Gypsum-only, P32 ×4 (n = 5)

Gypsum-only, P32 ×6 (n = 5)

Gypsum-only, P32 ×8 (n = 5)

Gypsum-only, P32 ×16 (n = 5)

Coupled gypsum–calcite, P32 ×4 (n = 5)

Coupled gypsum–calcite, P32 ×6 (n = 5)

Coupled gypsum–calcite, P32 ×8 (n = 5)

Coupled gypsum–calcite, P32 ×16 (n = 5)

Coupled ÷ gypsum-only, paired (n = 5)

Coupled ÷ gypsum-only, paired (n = 5)

Coupled ÷ gypsum-only, paired (n = 5)

Coupled ÷ gypsum-only, paired (n = 5)

fraction of the gypsum dissolved

0.0

0.2

0.4

0.6

0.8

0

200

400

600

800

1000

distance of fracture centre from inlet (m)

participation ratio PR_V, bin value / its start
(blue = flow concentrates, red = spreads)

3×10^0

2×10^0

10^0

6×10^{-1}

4×10^{-1}

coupled ÷ gypsum-only
(orange = coupled more spread)

2×10^0

10^0

6×10^{-1}

Participation ratio PR_V by distance from the inlet — Phase 1, transmissivity σ = 1

Each cell: fractures whose centre lies in that 100 m bin, median over 5 seeds. Solid line: median dissolved fraction at the network PR_V minimum; dashed: at breakthrough.

Gypsum-only, P32 ×4 (n = 5)

Gypsum-only, P32 ×6 (n = 5)

Gypsum-only, P32 ×8 (n = 5)

Gypsum-only, P32 ×16 (n = 5)

Coupled gypsum–calcite, P32 ×4 (n = 5)

Coupled gypsum–calcite, P32 ×6 (n = 5)

Coupled gypsum–calcite, P32 ×8 (n = 5)

Coupled gypsum–calcite, P32 ×16 (n = 5)

Coupled ÷ gypsum-only, paired (n = 5)

Coupled ÷ gypsum-only, paired (n = 5)

Coupled ÷ gypsum-only, paired (n = 5)

Coupled ÷ gypsum-only, paired (n = 5)

fraction of the gypsum dissolved

0.0

0.2

0.4

0.6

0.8

0

200

400

600

800

1000

distance of fracture centre from inlet (m)

participation ratio PR_V, bin value / its start
(blue = flow concentrates, red = spreads)

3×10^0

2×10^0

10^0

6×10^{-1}

4×10^{-1}

coupled ÷ gypsum-only
(orange = coupled more spread)

2×10^0

10^0

6×10^{-1}